



TrueCrypt

Learn the Art of Encryption

When did you last feel that you ought to have secured that important file so nobody else could view it without your permission? Ever lost your pen drive with loads of family photos or important spreadsheets? Do you often worry that Big Brother may be watching your uploads to the cloud? If the answer to all those questions is a “Yes,” the solution is encryption! Read on...

Many commercial encryption products can protect files, folders and partitions. Probably the most discussed is ‘Bitlocker’, which is bundled free with MS Windows 7 Ultimate. And if you have ever wondered if there is open source encryption software – there is and it’s called TrueCrypt.

As mentioned on *TrueCrypt.org*, it is a software system to establish and maintain an on-the-fly-encrypted volume (data storage device). On-the-fly implies that data is automatically encrypted right before it is saved, and decrypted right after it is read, without user intervention. No data stored on an encrypted volume can be decrypted without the correct password/keyfile(s) or correct encryption keys. The entire file system is encrypted.

Files can be copied to and from a mounted TrueCrypt volume just like with any normal disk (for example, by simple drag-and-drop operations). The software is just great, works as expected, is simple, and has a GUI for ease of use. Apart from these properties, I would say that its most important feature is the security TrueCrypt offers. It’s so secure that even FBI hackers failed to recover data from TrueCrypt encrypted volumes (see *References* for a link). Surprised? Here are some more facts: it is available under the GNU licence, works on almost all operating systems, including Linux, Mac OS and MS Windows 2000, and later versions.

Continue reading at your own risk... you may soon fall in love with TrueCrypt!

The basics of encryption

So let’s take a quick look at the basics of encryption. Everybody knows what a password is, and how it operates. Anybody can visualize encryption key to be same as the password. However,

technically, there are a few differences. Look at Table 1 for a comparison.

Table 1: Passwords and encryption keys

Password	Encryption key
Typically, a combination of a few characters to be entered before gaining access.	A combination of characters, generally slang or a pass-phrase or random character sequence required to decrypt the message.
Does not change the contents of the original resource.	Is used to actually change the data by using encryption techniques.
If lost, protection can typically be removed with ready-made tools such as spreadsheet and document password crackers/removers, to reveal the actual data.	It is practically impossible to retrieve (decrypt) data from an encrypted file unless the key is known. The only way is brute force, which can also be ineffective for a key with sufficient strength. (Be careful—lose the encryption key, and you lose all the data!)

Basically, there are three encryption mechanisms: the symmetric key, asymmetric key and hashing. For this article, we need to understand two of them.

Symmetric key encryption converts data from clear text (original format) to cipher text (encrypted format) using an encryption algorithm and an encryption key. The same key is